Dino v0.0.2 — Bus Transceiver Integration (74LS645)

Objective

Introduce bidirectional bus arbitration using the 74LS645 octal bus transceiver. Prevent bus contention between DIP switch input, SRAM, and LED output lines. Cleanly switch bus direction based on system mode (LOAD vs RUN).

New Component

Qty	Part	Description
1	74LS645N	Octal Bus Transceiver (3-state, DIR + /OE)

Updated Bus Architecture

Bus Participants

Device	Bus Direction	Mode
DIP Switch	Output to Bus	LOAD
SRAM	Bi-directional	LOAD & RUN
LED Display	Input from Bus	RUN

Arbitration Logic (Simplified Truth Table)

Mode	OE (74LS645)	DIR (74LS645)	SRAM /OE	SRAM /WE	Description
LOAD	LOW	HIGH	HIGH	LOW	DIP → SRAM write
RUN	LOW	LOW	LOW	HIGH	SRAM → LED read
FLOAT	HIGH	X	X	X	Bus idle (tri-stated)

Build Instructions: Modifications

Step 2: Modify SRAM Wiring

1. Insert 74LS645 between DIP switch and shared data bus:
   - A-side (DIP input) → DIP switch lines
   - B-side (Data Bus) → SRAM D0–D7 and LED bank
2. Connect DIR pin to system mode logic:
   - HIGH during LOAD
   - LOW during RUN

3. Connect OE pin to system mode logic:
    - LOW to enable transceiver
    - HIGH to tri-state outputs (FLOAT mode)

Step 3: Update Mode Switch

- Extend existing SPDT switch or replace with a control logic block (e.g., 2-bit selector or additional gates) to toggle DIR and OE accordingly.

Completion Criteria

- DIP switch drives data only during LOAD mode
- SRAM output drives bus only during RUN mode
- LED bank reads cleanly from SRAM with no back-driving
- DIP and LED banks are isolated from each other using 74LS645
- No overlapping drivers on shared 8-bit data bus
